

EECS 498 · AASE · FALL 2026

From functions to agent actions

Lab02 · September 21–22, 2026

40-minute example · 70-minute project work

The stockroom problem

“Prepare supplies for five workshop participants.”

Three markers available.

Two possible designs

- A complete plan first
- One decision after each observation
- User input at consequential choices

The design contract

- Model and application responsibilities
- Action inputs and outcomes
- User decisions and stopping conditions

Classification and agency

Classification

Items → one call → validated labels

Agent

Goal → action → observation → decision

Function and action contracts

```
# Application function
inventory.reserve(item, quantity)

# Model request
{"action": "reserve",
 "args": {"item": "marker", "quantity": 5}}
```

The feedback increment

- Read the increment spec
- Predict the failing test
- Implement with scoped Aider context
- Inspect the diff and result

The next decision

```
reserve(marker, 5)  
→ only 3 available  
ask whether participants can share  
→ user agrees  
reserve(marker, 3)  
→ 3 reserved
```


Evidence in application state

- Three markers reserved
- Rejection leaves stock unchanged
- Repeated failures stop

A defensible submission

- Reasoned design choices
- Working action and observation flow
- Evidence and acknowledged limitations

Your project increment

- One observable result
- Relevant spec and context
- A test that detects failure

Design reconciliation

- Update decisions and diagrams
- Record evidence and limitations
- Commit and push